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COURSE CODE: SBT-DF203-BASIC NETWORKING SKILLS

LAB NAME: LAB5 ARP POISONING FORENSICS

DATE: 17/09/26

INSTRUCTOR: AMINU IDRIS

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 5: ARP Poisoning Forensics

Three-Week Delivery Block 2/3 | 12-18 September 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction

Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 5
Lab Title	ARP Poisoning Forensics
Delivery Block	2/3 of 3
Scheduled Dates	12-18 September 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	arp.pcap and an isolated host-only ARP capture

1. Source Materials and Slide Alignment

- Address Resolution Protocol Forensics: ARP operation, ARP caches, request/reply fields, capture with tshark, ARP spoofing/poisoning, man-in-the-middle consequences and evidence analysis.
- The slide practical compares clean and poisoned ARP tables and uses a supplied arp.py script only within the same LAN.
- The assignment requires legitimate ARP capture, spoofing capture and proof of a poisoned ARP table.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A user reports that the default gateway intermittently resolves to the wrong MAC address. You will first document normal ARP resolution, then analyze a known ARP-poisoning capture. In an instructor-approved host-only network, you may reproduce a short controlled poisoning event, capture the evidence, restore the ARP mappings and explain how the attack is detected.

3. Learning Outcomes

- Explain how ARP maps IPv4 addresses to MAC addresses on a LAN.
- Inspect ARP caches on Linux and Windows systems.
- Identify ARP requests, legitimate replies and unsolicited or conflicting replies.
- Detect one IP address being associated with multiple MAC addresses.
- Capture and document a bounded ARP-poisoning simulation in a host-only network.
- Restore network state and recommend preventive controls.

4. Legal, Ethical and Safety Rules

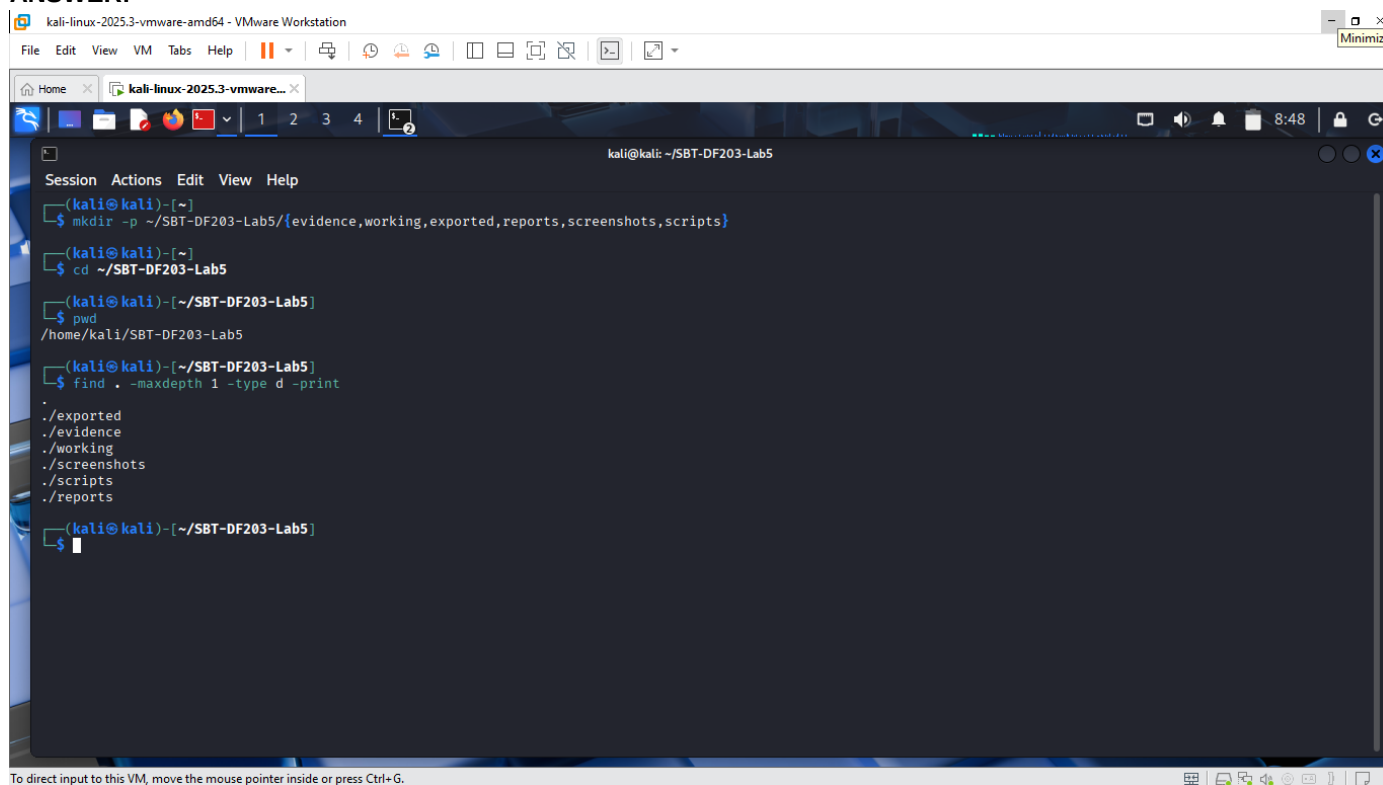
- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- The live component is optional and must use instructor-assigned RFC1918 addresses on a host-only/internal virtual switch. Never poison a home, school, office, hotel or public Wi-Fi network.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Virtual machines	Kali attacker/analyst plus Linux or Windows victim	Create an isolated LAN scenario.
Network mode	Host-only/internal network	Prevent access to production networks.
Capture tools	Wireshark and tshark	Capture ARP packets.
Packet library	python3-scapy	Run authorized training scripts.
System tools	ip neigh, arp, ipconfig /all	Inspect ARP tables and interfaces.
Evidence	arp.pcap or student capture	Analyze known poisoning indicators.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab5/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab5
pwd
find . -maxdepth 1 -type d -print
```

ANSWER:

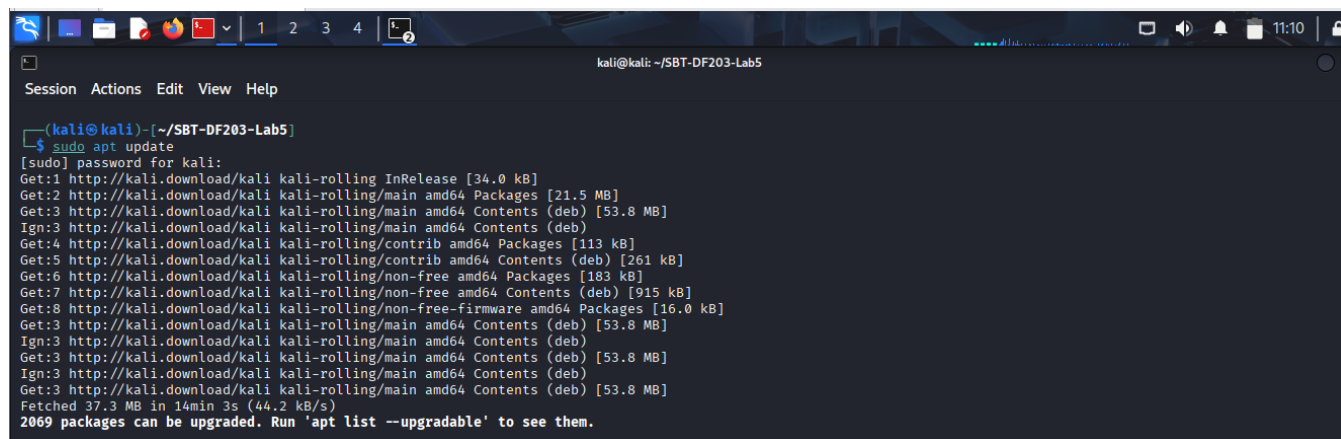
Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```

sudo apt update
sudo apt install -y wireshark tshark python3-scapy net-tools
wget -O evidence/arp.pcap \
  'https://raw.githubusercontent.com/frankwxu/digital-forensics-
  lab/main/Networking_Forensics/lab_files/ARP_spoofing/arp.pcap'
cp --preserve=timestamps evidence/arp.pcap working/arp_working.pcap
sha256sum evidence/arp.pcap working/arp_working.pcap | tee reports/arp_capture_hashes.txt
ip -br address | tee reports/interfaces.txt
ip route | tee reports/routes.txt
ip neigh show | tee reports/arp_table_initial.txt

```

Screenshot required: Interface addresses, route/gateway, initial ARP table and evidence hashes.

ANSWER:

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~/SBT-DF203-Lab5

Session Actions Edit View Help
(kali@kali)~/SBT-DF203-Lab5
$ sudo apt install -y wireshark tshark python3-scapy net-tools
[sudo] password for kali:
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
python3-scapy is already the newest version (2.7.0+dfsg1-1).
net-tools is already the newest version (2.10-2).
The following packages were automatically installed and are no longer required:
  amass-common libbson-1.0-0t64 libplacebo349 oracle-instantclient-basic python3-tld
  binutils-mingw-w64-i686 libcrypt-dev libportmidi0 python3-bluepy python3-yaswfp
  binutils-mingw-w64-x86_64 libgavl1 libpython3.13-dev python3-click-plugins python3-zombie-imp
  dnsmasq libgdal37 libraw1e0.7 python3-fs python3.13 python3.13-dev
  dsniff libgirepository-1.0-1 libraw23t64 python3-gpg python3.13-minimal
  ettercap-common libimath-3-1-29t64 librsmb2-6 python3-gpg python3.13-venv
  ettercap-graphical libjs-jquery-ui libtheoraec1 python3-kismetcapturebtgeiger rpcsvc-proto
  figlet libkmlbase1t64 libudfread0 python3-kismetcapturefreaklabszigbee rsh-redone-client
  finger libkmlengine1t64 libwiretap15 python3-kismetcaptureertl433 samba-ad-dc
  gcc-mingw-w64-base libkmldev1t64 libwiresniff python3-kismetcaptureertladsb samba-ad-provision
  gcc-mingw-w64-i686-win32 liblua5.1-2 libwsutil16 python3-kismetcaptureertlamr samba-dsdb-modules
  gcc-mingw-w64-x86_64-win32 liblua5.1-common libxml2 python3-pefile smtp-user-enum
  gcc-mingw-w64-x86_64-win32-runtime libmbedcrypto16 libxml2 python3-pyexploitdb sparta-scripts
  girl1.2-girepository-2.0 libmjpegutils-2.1-0t64 libyelp0 python3-pyfiglet toilet-fonts
  libaio1t64 libmongoc-1.0-0t64 libzxing3 python3-pyshodan unicornscan
  libarmadillo14 libmpeg2encpp-2.1-0t64 medusa python3-qasync python3-qrcode urlscan
  libavc1394-0 libblosc2-4 libmpeg2-2.1-0t64 mingw-w64-common python3-qrcode python3-serial-asyncio wapiti
  libbluray2 libnids1.2t64 mingw-w64-i686-dev python3-smmmap
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2069
(kali@kali)~/SBT-DF203-Lab5
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~/SBT-DF203-Lab5

Session Actions Edit View Help
(kali@kali)~/SBT-DF203-Lab5
$ wget -O evidence/arp.pcap \
  'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/ARP_spoofing/arp.pcap'

--2026-09-15 11:21:43-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/ARP_spoofing/arp.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.110.133, 185.199.109.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)[185.199.110.133]:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1072 (1.0K) [application/octet-stream]
Saving to: 'evidence/arp.pcap'

evidence/arp.pcap      100%[=====] 1.05K --.-KB/s in 0s

2026-09-15 11:21:44 (6.48 MB/s) - 'evidence/arp.pcap' saved [1072/1072]

(kali@kali)~/SBT-DF203-Lab5
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

kali@kali: ~/SBT-DF203-Lab5
$ cp --preserve-timestamps evidence/arp.pcap working/arp_working.pcap

kali@kali: ~/SBT-DF203-Lab5
$ sha256sum evidence/arp.pcap working/arp_working.pcap | tee reports/arp_capture_hashes.txt
342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e  evidence/arp.pcap
342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e  working/arp_working.pcap

kali@kali: ~/SBT-DF203-Lab5
$ ip -br address | tee reports/interfaces.txt
lo                UNKNOWN    127.0.0.1/8  ::1/128
eth0              UP        192.168.242.130/24  fe80::fe8d:6b5d:6ee4:ee1b/64
docker0          DOWN     172.17.0.1/16

kali@kali: ~/SBT-DF203-Lab5
$ ip route | tee reports/routes.txt
default via 192.168.242.2 dev eth0 proto dhcp src 192.168.242.130 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.242.0/24 dev eth0 proto kernel scope link src 192.168.242.130 metric 100

kali@kali: ~/SBT-DF203-Lab5
$ ip neigh show | tee reports/arp_table_initial.txt
192.168.242.254 dev eth0 lladdr 00:50:56:fe:21:82 STALE
192.168.242.2 dev eth0 lladdr 00:50:56:e5:bb:77 STALE

kali@kali: ~/SBT-DF203-Lab5
$

```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab5-ADOYI NATHANIEL JOHNSON
Trainee name	ADOYI NATHANIEL JOHNSON
Date and time started	15/09/26 12:30PM
Evidence file name(s)	SBT-DF203_LAB5_ARP POISONING FORENSICS
Source or generation method	KALI COMMAND LINE
Original SHA-256	342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e evidence/arp.pcap
Working-copy SHA-256	342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e working/arp_working.pcap
Analysis workstation/VM	KALI@KALI
Notes on any changes	NIL

8. Part A - Observe Normal ARP Resolution

1. Identify the victim IP, gateway IP and your interface name.
2. Clear only the specific training neighbor entry where permitted.
3. Start an ARP-only capture, ping the gateway once and stop the capture.
4. Identify the broadcast request and unicast reply.

```

IFACE=eth0
GATEWAY_IP=$(ip route | awk '/default/ {print $3; exit}')
sudo ip neigh flush "$GATEWAY_IP" dev "$IFACE"
sudo tshark -i "$IFACE" -f 'arp' -a duration:20 -w evidence/normal_arp.pcapng &
sleep 2
ping -c 1 "$GATEWAY_IP"
wait
ip neigh show | tee reports/arp_table_after_ping.txt

```

ANSWER:

```

kali@kali: ~/SBT-DF203-Lab5
$ IFACE=eth0
$ GATEWAY_IP=$(ip route | awk '/default/ {print $3; exit}')
$ sudo ip neigh flush "$GATEWAY_IP" dev "$IFACE"
[sudo] password for kali:
$ sudo tshark -i "$IFACE" -f 'arp' -a duration:20 -w evidence/normal_arp.pcapng &
sleep 2
[1] 234227
$ ping -c 1 "$GATEWAY_IP"
wait
PING 192.168.242.2 (192.168.242.2) 56(84) bytes of data:
64 bytes from 192.168.242.2: icmp_seq=1 ttl=128 time=4.90 ms

--- 192.168.242.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 4.895/4.895/4.895/0.000 ms
$ ip neigh show | tee reports/arp_table_after_ping.txt
192.168.242.254 dev eth0 lladdr 00:50:56:fe:21:82 STALE
192.168.242.2 dev eth0 lladdr 00:50:56:e5:bb:77 REACHABLE

```

9. Part B - Analyze ARP Request and Reply Fields

```

tshark -r evidence/normal_arp.pcapng -Y 'arp' -T fields \
-e frame.number -e frame.time -e eth.src -e eth.dst -e arp.opcode \
-e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/normal_arp_fields.tsv

```

Field	Normal Request	Normal Reply
Ethernet destination	ff:ff:ff:ff:ff:ff	Requester MAC
ARP opcode	1	2
Sender protocol address	Requester IP	Target IP
Sender hardware address	Requester MAC	Target MAC
Target protocol address	Target IP	Requester IP
Forensic interpretation	Who has target IP?	Target IP is at target MAC.

ANSWER:

The screenshot shows a Kali Linux terminal window titled "kali-linux-2025.3-vmware-amd64 - VMware Workstation". The terminal prompt is "(kali@kali)~[~/SBT-DF203-Lab5]". The user enters the following command:

```
tshark -r evidence/normal_arp.pcapng -Y 'arp' -T fields \
-e frame.number -e frame.time -e eth.src -e eth.dst -e arp.opcode \
-e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/normal_arp_fields.tsv
```

The terminal output shows a warning: "Warning: program compiled against libxml 215 using older 214" and an error: "tshark: The file 'evidence/normal_arp.pcapng' doesn't exist." The terminal prompt returns to "(kali@kali)~[~/SBT-DF203-Lab5]".

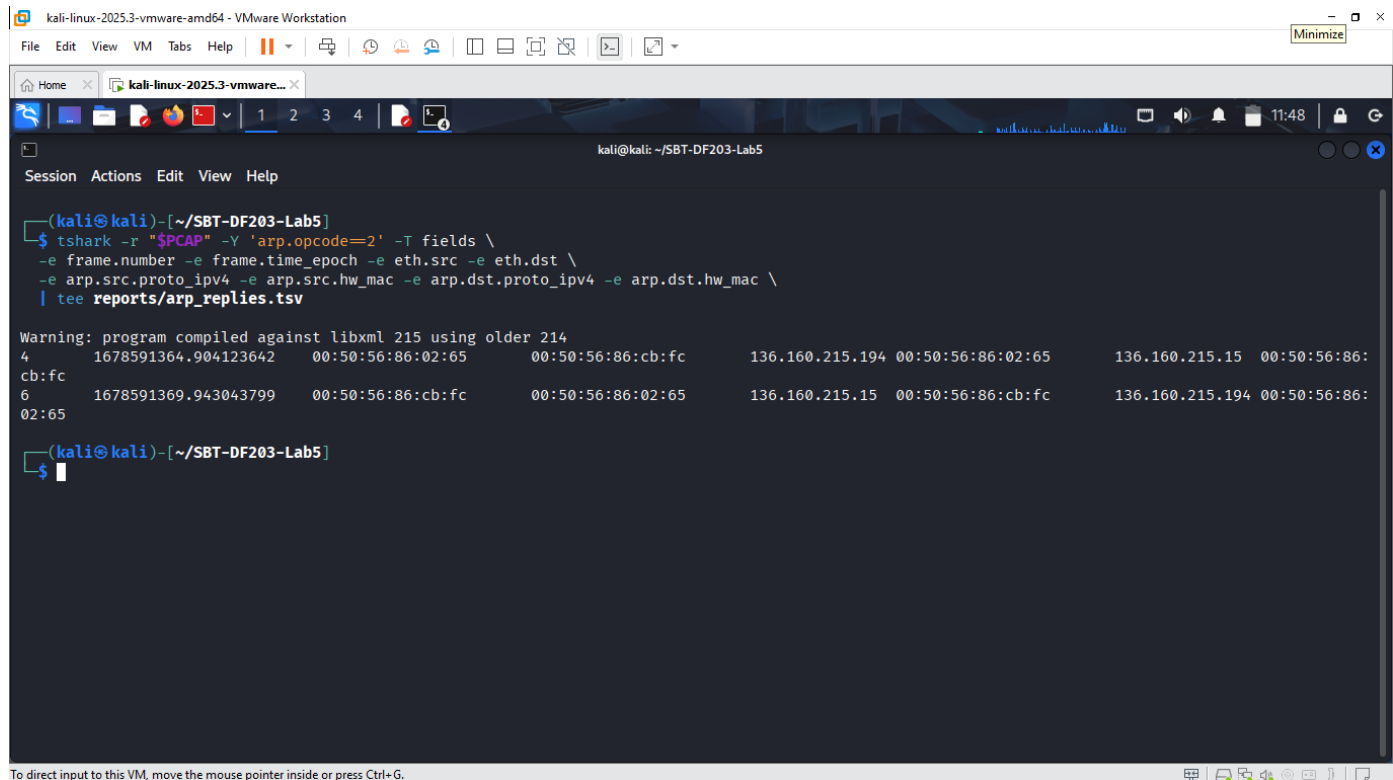
10. Part C - Analyze the Supplied Poisoning Capture

```
PCAP=working/arp_working.pcap
# Inventory all ARP claims
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst \
-e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
| tee reports/arp_replies.tsv

# Summarize IP-to-MAC claims
tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields -e arp.src.proto_ipv4 -e arp.src.hw_mac \
| sort | uniq -c | sort -nr | tee reports/ip_mac_claims.txt

# Look for gratuitous or unsolicited-looking replies
tshark -r "$PCAP" -Y 'arp.opcode==2 && eth.dst!=ff:ff:ff:ff:ff:ff' -T fields \
-e frame.number -e frame.time -e arp.src.proto_ipv4 -e arp.src.hw_mac -e eth.dst \
| tee reports/unicast_arp_replies.tsv
```

ANSWER:

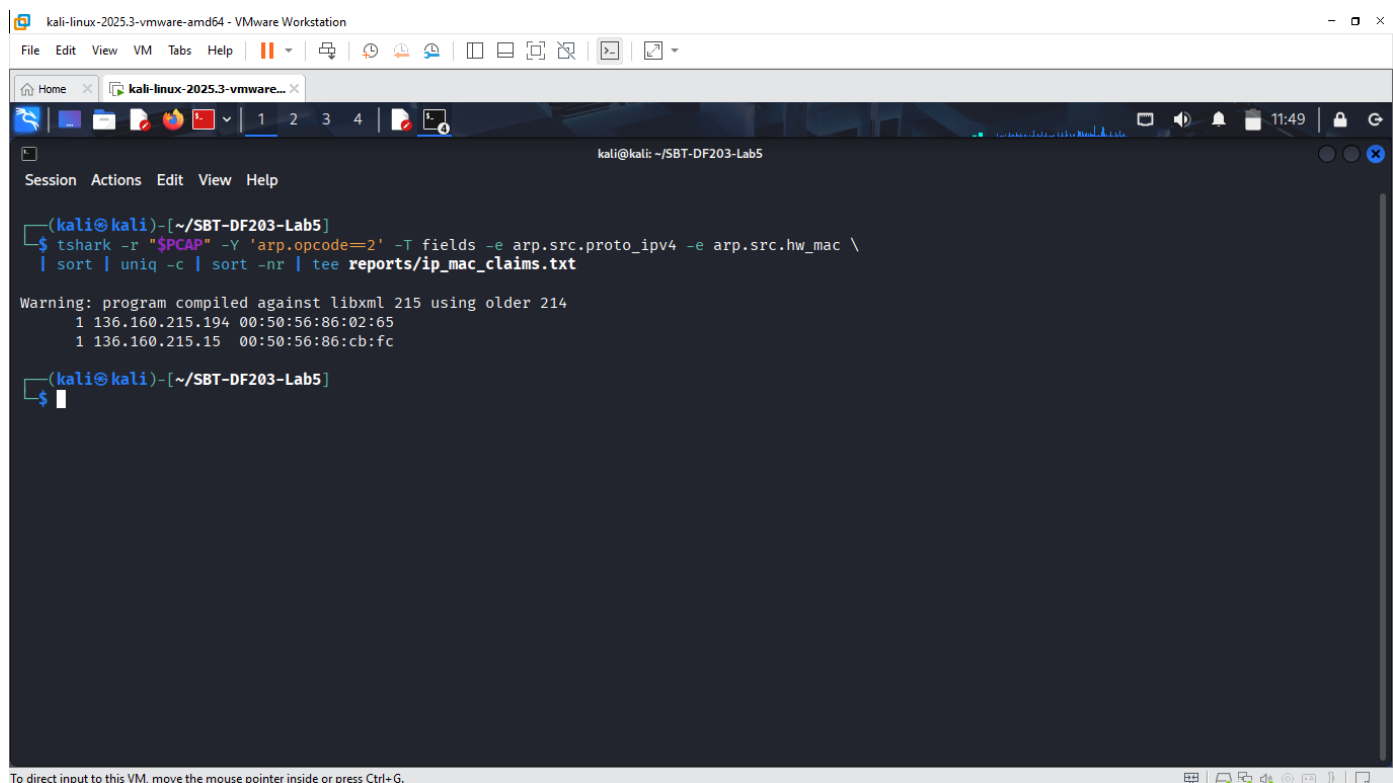


```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
kali@kali: ~/SBT-DF203-Lab5
Session Actions Edit View Help
(kali@kali)~[~/SBT-DF203-Lab5]
$ tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst \
  -e arp.src.proto_ipv4 -e arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac \
  | tee reports/arp_replies.tsv

Warning: program compiled against libxml 215 using older 214
4      1678591364.904123642      00:50:56:86:02:65      00:50:56:86:cb:fc      136.160.215.194 00:50:56:86:02:65      136.160.215.15 00:50:56:86:cb:fc
cb:fc
6      1678591369.943043799      00:50:56:86:cb:fc      00:50:56:86:02:65      136.160.215.15 00:50:56:86:cb:fc      136.160.215.194 00:50:56:86:02:65
02:65

(kali@kali)~[~/SBT-DF203-Lab5]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
kali@kali: ~/SBT-DF203-Lab5
Session Actions Edit View Help
(kali@kali)~[~/SBT-DF203-Lab5]
$ tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields -e arp.src.proto_ipv4 -e arp.src.hw_mac \
  | sort | uniq -c | sort -nr | tee reports/ip_mac_claims.txt

Warning: program compiled against libxml 215 using older 214
1 136.160.215.194 00:50:56:86:02:65
1 136.160.215.15 00:50:56:86:cb:fc

(kali@kali)~[~/SBT-DF203-Lab5]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

(kali@kali)~/SBT-DF203-Lab5
$ tshark -r "$PCAP" -Y 'arp.opcode==2 && eth.dst!=ff:ff:ff:ff:ff:ff' -T fields \
-e frame.number -e frame.time -e arp.src.proto_ipv4 -e arp.src.hw_mac -e eth.dst \
| tee reports/unicast_arp_replies.tsv

Warning: program compiled against libxml 215 using older 214
4      2023-03-11T22:22:44.904123642-0500      136.160.215.194  00:50:56:86:02:65      00:50:56:86:cb:fc
6      2023-03-11T22:22:49.943043799-0500      136.160.215.15  00:50:56:86:cb:fc      00:50:56:86:02:65

(kali@kali)~/SBT-DF203-Lab5
$

```

11. Part D - Optional Controlled Host-Only Simulation

Instructor Approval Required

Skip this section unless the virtual network is confirmed as host-only/internal and the victim, gateway and analyst IP addresses are assigned specifically for the lab.

- Record clean ARP tables on victim and gateway.
- Start ARP capture on the analyst VM.
- Run the instructor-provided arp.py for no more than 30 seconds.
- Capture the changed victim mapping and stop the script.
- Restore correct ARP mappings or restart the isolated VMs.

Example placeholders only - replace with instructor-assigned lab IPs

VICTIM_IP=192.168.56.20

GATEWAY_IP=192.168.56.1

IFACE=eth0

```
sudo tshark -i "$IFACE" -f 'arp' -a duration:40 -w evidence/controlled_arp_poison.pcapng &
sleep 2
```

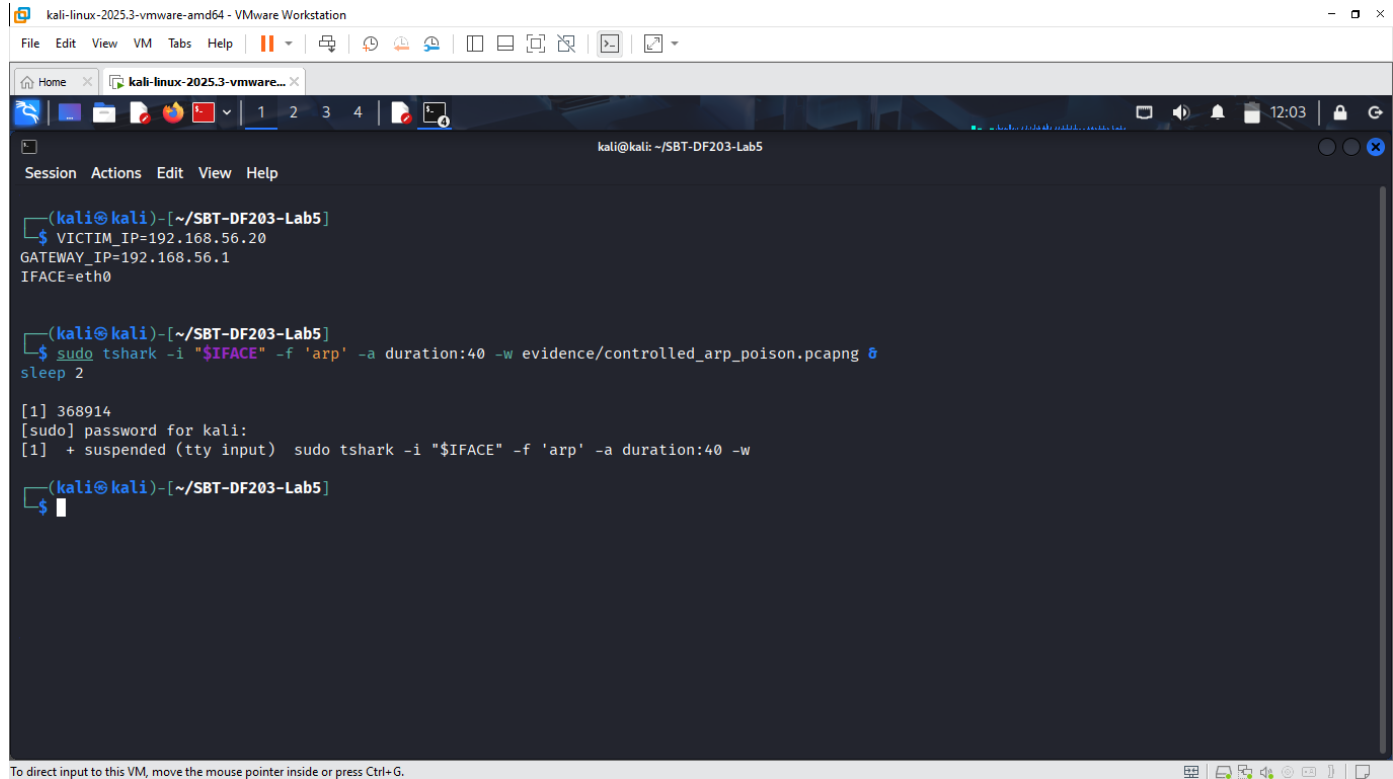
In another terminal, run the instructor-provided script for <=30 seconds:

```
# sudo timeout 30 python3 scripts/arp.py "$VICTIM_IP" "$GATEWAY_IP"
```

wait

```
ip neigh show | tee reports/analyst_arp_table_after.txt
```

ANSWER:

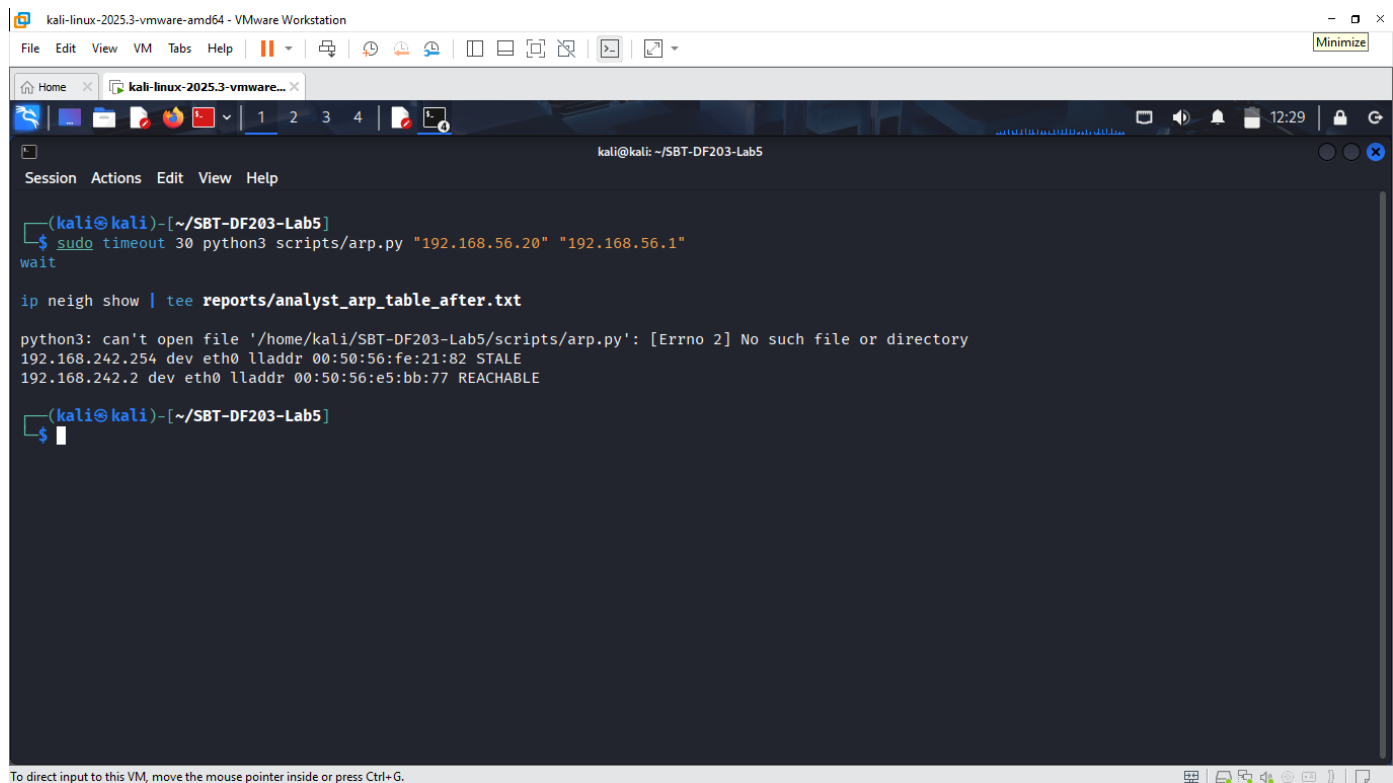


```
kali@kali: ~/SBT-DF203-Lab5
(kali@kali)~-[~/SBT-DF203-Lab5]
$ VICTIM_IP=192.168.56.20
GATEWAY_IP=192.168.56.1
IFACE=eth0

(kali@kali)~-[~/SBT-DF203-Lab5]
$ sudo tshark -i "$IFACE" -f 'arp' -a duration:40 -w evidence/controlled_arp_poison.pcapng &
sleep 2

[1] 368914
[sudo] password for kali:
[1] + suspended (tty input) sudo tshark -i "$IFACE" -f 'arp' -a duration:40 -w

(kali@kali)~-[~/SBT-DF203-Lab5]
$
```



```
kali@kali: ~/SBT-DF203-Lab5
(kali@kali)~-[~/SBT-DF203-Lab5]
$ sudo timeout 30 python3 scripts/arp.py "192.168.56.20" "192.168.56.1"
wait

ip neigh show | tee reports/analyst_arp_table_after.txt

python3: can't open file '/home/kali/SBT-DF203-Lab5/scripts/arp.py': [Errno 2] No such file or directory
192.168.242.254 dev eth0 lladdr 00:50:56:fe:21:82 STALE
192.168.242.2 dev eth0 lladdr 00:50:56:e5:bb:77 REACHABLE

(kali@kali)~-[~/SBT-DF203-Lab5]
$
```

```

kali@kali: ~/SBT-DF203-Lab5
$ ip neigh show | tee reports/analyst_arp_table_after.txt
192.168.242.254 dev eth0 lladdr 00:50:56:fe:21:82 STALE
192.168.242.2 dev eth0 lladdr 00:50:56:e5:bb:77 REACHABLE

```

12. Part E - Detection Logic and Timeline

Time	Claimed IP	Claimed MAC	Target	Request Seen First?	Assessment
05:15:29	136.160.215.194	00:50:56:86:02:65	00:50:56:86:CB:FC		
05:15:35	136.160.215.15	00:50:56:86:CB:CF	00:50:56:86:02:65		
	192.168.242.254	00:50:56:fe:21:82	00:50:56:fe:21:82		STALE
	192.168.242.2	00:50:56:e5:bb:77	00:50:56:e5:bb:77		REACHABLE

- Flag a gateway IP that changes from its legitimate MAC to the analyst/attacker MAC.
- Flag repeated ARP replies without a corresponding recent request.
- Flag one MAC claiming both victim and gateway IP addresses.
- Correlate ARP evidence with packet forwarding, DNS/HTTP anomalies and endpoint ARP-cache changes.

13. Restoration and Prevention

```

# Stop any training script first, then clear learned entries on the lab VM
sudo ip neigh flush all
# Renew/reconnect the isolated interface as appropriate
ip neigh show

# Confirm no ARP-poisoning process remains
ps aux | grep -E '[a]rp.py|[s]copy' | tee reports/process_check.txt

```

ANSWER:

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
kali@kali: ~/SBT-DF203-Lab5
Session Actions Edit View Help
(kali@kali)~/SBT-DF203-Lab5
$ sudo ip neigh flush all
(kali@kali)~/SBT-DF203-Lab5
$ ip neigh show
192.168.242.2 dev eth0 lladdr 00:50:56:e5:bb:77 REACHABLE
(kali@kali)~/SBT-DF203-Lab5
$ ps aux | grep -E '[a]rp.py|[s]copy' | tee reports/process_check.txt
(kali@kali)~/SBT-DF203-Lab5
$

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

- Use Dynamic ARP Inspection with DHCP snooping on managed switches where available.
- Segment networks, enable port security and monitor gateway IP-to-MAC consistency.
- Use encrypted application protocols so interception does not expose plaintext.
- Deploy ARP monitoring tools and correlate alerts with switch and endpoint logs.

Required Screenshots Checklist

- ☐ Initial interface, route and ARP table.
- ☐ Normal broadcast ARP request.
- ☐ Normal ARP reply.
- ☐ Supplied capture hash.
- ☐ ARP reply inventory.
- ☐ IP-to-MAC claim summary.
- ☐ Conflicting gateway MAC evidence.
- ☐ Clean versus poisoned ARP table comparison.
- ☐ Controlled capture, if authorized.
- ☐ Restored ARP table and process check.

Student Report Template

10. Executive summary and authorization.
11. Network diagram and assigned IP/MAC addresses.
12. Evidence integrity.
13. Normal ARP behaviour.
14. Poisoning indicators and timeline.
15. ARP-table comparison.
16. Impact assessment.
17. Restoration and preventive controls.
18. Appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab5_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Safety and environment	15	Host-only scope and restoration documented.
Normal ARP analysis	15	Request/reply fields and ARP-cache behaviour.
Poisoning detection	25	Conflicting claims, unsolicited replies and timeline.
Controlled evidence or supplied PCAP analysis	20	Complete, repeatable capture analysis.
Mitigation and restoration	15	Correct cleanup and preventive controls.
Report quality	10	Clear diagram, screenshots and conclusions.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
ARP request opcode	1.
ARP reply opcode	2.
Broadcast destination	ff:ff:ff:ff:ff:ff for a typical request.
Poisoning indicator	An IP, especially the gateway, is claimed by a different MAC.
Unsolicited reply	ARP reply appears without a corresponding request.
MITM pattern	Attacker MAC is inserted for victim and/or gateway IP mappings.

Command Reference Summary

Command	Purpose
ip neigh show	Display Linux neighbor/ARP cache.
arp -a	Display legacy ARP table.
arp.opcode==1	Filter ARP requests.
arp.opcode==2	Filter ARP replies.
arp.src.proto_ipv4	Claimed sender IP.
arp.src.hw_mac	Claimed sender MAC.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.